INCH-POUND

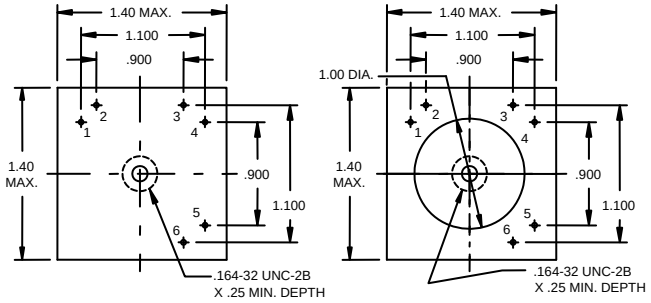
MIL-PRF-27/47C
9 November 2009
SUPERSEDING
MIL-PRF-27/47B
7 February 1984

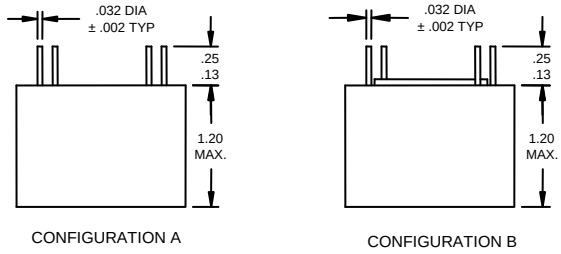
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 25 VOLT AMPERES, 400 HZ ELECTROSTATIC SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification sheet and MIL-PRF-27.

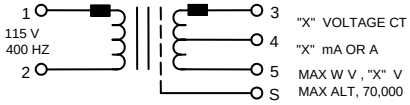




CONFIGURATION A

CONFIGURATION B

CIRCUIT DIAGRAM AND MARKINGS



(SEE TABLE I FOR VALUE OF "X")

Inches	mm
.002	0.05
.02	0.5
.03	0.8
.032	0.81
.13	3.3
.164	.17
.25	6.4
.900	22.86
1.00	25.4
1.10	27.9
1.20	30.5
1.40	35.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm).
4. Marking shall be on the side of the case.
5. A white dot shall be marked on the side opposite the terminals, over terminal 1, in addition to the terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:18Z
Check the source to verify that this is the current version before use.

REQUIREMENTS: (When numbers in parentheses, such as (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 4.4 ounces, maximum.

Duty cycle: Continuous.

Case: Plastic encapsulated.

Material: Plastic.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55˚C to +130˚C, maximum.

Terminals: Printed circuit pins - tinned brass.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts rms, 400 Hz across (1-2): Current in (1-2) shall not exceed 25 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a restrictive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 88 percent minimum at rated load.

Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{\text{Voltage (no Load) - Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$

shall not exceed 7 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 45˚C, maximum.

Vibration: MIL-STD-202, method 204, test condition D.

Marking location: See figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:18Z
Check the source to verify that this is the current version before use.

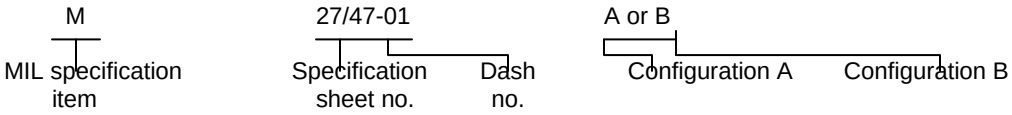
VERIFICATION

Extent of qualification

Qualification testing and approval to M27/37-44 A or B, M27/42-44 A or B, and M27/46-44 A or B shall be sufficient to grant qualification approval to MIL-PRF-27/37 through MIL-PRF-27/47 inclusive, all parts.

Qualification testing and approval to M27/47-44 shall be sufficient to grant qualification approval to M27/47-01 through M27/47-44.

Part or Identifying Number (PIN), example:



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:18Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical ratings.

Dash No.	Type designation	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms) (MAX)
01 A or B	840	6	4.167	325
02 A or B	841	6.3	3.968	325
03 A or B	842	7	3.571	325
04 A or B	843	7.5	3.333	325
05 A or B	844	8	3.125	325
06 A or B	845	9	2.778	325
07 A or B	846	10	2.500	325
08 A or B	847	11	2.273	325
09 A or B	848	12	2.083	325
10 A or B	849	12.6	1.984	325
11 A or B	850	13	1.923	325
12 A or B	851	15	1.667	325
13 A or B	852	16	1.563	325
14 A or B	853	18	1.389	325
15 A or B	854	20	1.250	325
16 A or B	855	22	1.136	325
17 A or B	856	24	1.042	325
18 A or B	857	26	0.962	325
19 A or B	858	28	0.893	325
20 A or B	859	30	0.833	325
21 A or B	860	33	0.758	325
22 A or B	861	36	0.694	325
23 A or B	862	39	0.641	325
24 A or B	863	43	0.581	325
25 A or B	864	47	0.532	325
26 A or B	865	51	0.490	325
27 A or B	866	56	0.446	325
28 A or B	867	62	0.403	325
29 A or B	868	68	0.367	325
30 A or B	869	75	0.333	325
31 A or B	870	82	0.305	325
32 A or B	871	91	0.275	325
33 A or B	872	100	0.250	325
34 A or B	873	110	0.227	325
35 A or B	874	115	0.218	325
36 A or B	875	120	0.208	325
37 A or B	876	130	0.192	325
38 A or B	877	150	0.166	325
39 A or B	878	180	0.139	325
40 A or B	879	200	0.125	440
41 A or B	880	220	0.113	440
42 A or B	881	240	0.104	440
43 A or B	882	270	0.093	580
44 A or B	883	300	0.083	580

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:18Z
Check the source to verify that this is the current version before use.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-PRF-27/37	MIL-PRF-27/38	MIL-PRF-27/39	MIL-PRF-27/40
MIL-PRF-27/41	MIL-PRF-27/42	MIL-PRF-27/43	MIL-PRF-27/44
MIL-PRF-27/45	MIL-PRF-27/46	MIL-STD-202	

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2009-042)

Review activities:
Army - CR4, MI
Navy - AS, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:18Z
Check the source to verify that this is the current version before use.